

Chapter 1



Data Analysis

Section 1.3

Describing Quantitative Data with Numbers



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Displaying Quantitative Data with Numbers

LEARNING TARGETS

By the end of this section, you should be able to:

- ✓ CALCULATE measures of center (mean, median) for a distribution of quantitative data.
- ✓ CALCULATE and INTERPRET measures of variability (range, standard deviation, IQR) for a distribution of quantitative data.
- ✓ EXPLAIN how outliers and skewness affect measures of center and variability.
- ✓ IDENTIFY outliers using the $1.5 \times \text{IQR}$ rule.
- ✓ MAKE and INTERPRET boxplots of quantitative data.
- ✓ Use boxplots and numerical summaries to COMPARE distributions of quantitative data.



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Measuring Center: The Mean

The **mean** of a distribution of quantitative data is the average of all the individual data values. To find the mean, add all the values and divide by the total number of observations.

If the n observations are $x_1, x_2, \dots, x_n$, the sample mean $\bar{x}$ (pronounced “x-bar”) is given by the following formula:

$$\bar{x} = \frac{\text{sum of data values}}{\text{number of data values}} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x_i}{n}$$



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Measuring Center: The Mean

Here are the data on the number of goals scored in 20 games played by the 2016 U.S. women’s soccer team:

5 5 1 10 5 2 1 1 2 3 3 2 1 4 2 1 2 1 9 3



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5 5 1 10 5 2 1 1 2 3 3 2 1 4 2 1 2 1 9 3

$$\bar{x} = \frac{1+1+1+1+1+1+2+2+2+2+2+3+3+3+4+5+5+5+9+10}{20}$$

$\bar{x} = 3.15 \text{ goals}$



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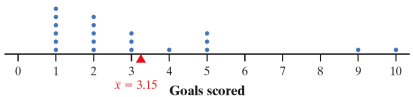
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$\bar{x} = 3.15 \text{ goals}$





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Measuring Center: The Mean

The symbol $\bar{x}$ refers to the mean of a *sample*. ← **Statistic**

The notation μ refers to the mean of a *population*. ← **Parameter**

A **statistic** is a number that describes some characteristic of a *sample*.

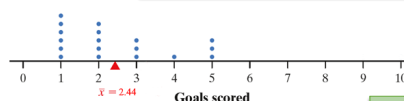
A **parameter** is a number that describes some characteristic of a *population*.

Measuring Center: The Mean

Here is the mean number of goals scored by the 2016 U.S. women's soccer team, **if we exclude the games that are possible outliers** (when they scored 9 and 10 goals).

$$\bar{x} = \frac{1+1+1+1+1+1+2+2+2+2+2+3+3+3+4+5+5+5}{18}$$

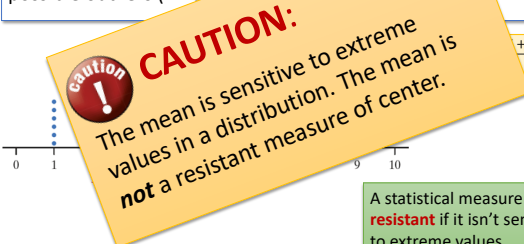
$$\bar{x} = 2.44 \text{ goals}$$



A statistical measure is **resistant** if it isn't sensitive to extreme values.

Measuring Center: The Mean

Here is the mean number of goals scored by the 2016 U.S. women's soccer team, if we exclude the games that are possible outliers (when they scored 9 and 10 goals).



A statistical measure is **resistant** if it isn't sensitive to extreme values.

Measuring Center: The Median

The **median** is the midpoint of a distribution, the number such that about half the observations are smaller and about half are larger.

To find the median, arrange the data values from smallest to largest.

- If the number n of data values is odd, the median is the middle value in the ordered list.
- If the number n of data values is even, the median is the average of the two middle values in the ordered list.

Measuring Center: The Median

Here are the highway fuel economy ratings for a sample of 25 model year 2018 Toyota 4Runners tested by the EPA:

Raw data

22.4 22.4 22.3 23.3 22.3 22.3 22.5 22.4 22.1 21.5 22.0 22.2 22.7
22.8 22.4 22.6 22.9 22.5 22.1 22.4 22.2 22.9 22.6 21.9 22.4

Sorted data

21.5 21.9 22.0 22.1 22.1 22.2 22.2 22.3 22.3 22.3 22.4 22.4 22.4
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Median

Measuring Center: The Median

Here are the data on the number of goals scored in 20 games played by the 2016 U.S. women's soccer team:

Raw data

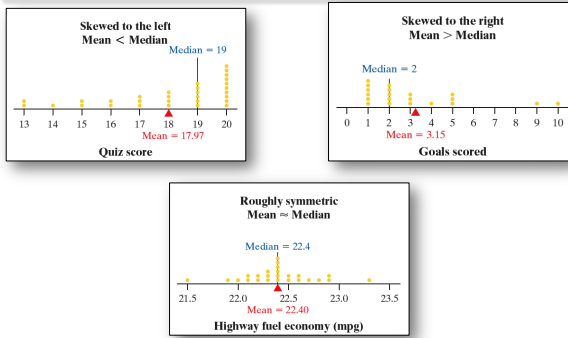
5 5 1 10 5 2 1 1 2 3 3 2 1 4 2 1 2 1 9 3

Sorted data

1 1 1 1 1 1 2 2 2 2 3 3 3 4 5 5 5 9 10

$$\text{Median} = \frac{2+2}{2} = 2$$

Comparing the Mean and the Median



Comparing the Mean and the Median

Effect of Skewness and Outliers on Measures of Center

- If a distribution of quantitative data is roughly symmetric and has no outliers, the mean and median will be similar.
- If the distribution is strongly skewed, the mean will be pulled in the direction of the skewness but the median won't. For a right-skewed distribution, we expect the mean to be greater than the median. For a left-skewed distribution, we expect the mean to be less than the median.
- The median is resistant to outliers but the mean isn't.

Measuring Variability: The Range

The **range** of a distribution is the distance between the minimum value and the maximum value. That is,
Range = Maximum – Minimum

Here are the data on the number of goals scored in 20 games played by the 2016 U.S. women's soccer team:

5 5 1 10 5 2 1 1 2 3 3 2 1 4 2 1 2 1 9 3

Range = 10 – 1 = 9 goals

CAUTION:

- The range of a data set is a single number.
- The range is **not** a resistant measure of variability.

Measuring Variability: The Standard Deviation

How to calculate standard deviation and variance:

- Find the mean of the distribution.
- Calculate the *deviation* (value – mean) of each value from the mean.
- Square each of the deviations.
- Add all the squared deviations, divide by $n - 1$. This is the **sample variance**.
- Take the square root. This is the **sample standard deviation**.

The **standard deviation** measures the typical distance of the values in a distribution from the mean.

$$s_x = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Measuring Variability: The Standard Deviation

Eleven high school students were asked how many "close" friends they have. Here are their responses: 1 2 2 2 3 3 3 3 4 4 6

How to calculate standard deviation, s_x :

- Find the mean of the distribution.

$$\bar{x} = \frac{1 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 4 + 4 + 6}{11} = 3$$

x_i
1
2
2
2
3
3
3
3
4
4
6

Measuring Variability: The Standard Deviation

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How to calculate standard deviation, s_x :

- Find the mean of the distribution.
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$$\bar{x} = \frac{1 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 4 + 4 + 6}{11} = 3$$

x_i	$x_i - \bar{x}$
1	1 - 3 = -2
2	2 - 3 = -1
2	2 - 3 = -1
2	2 - 3 = -1
3	3 - 3 = 0
3	3 - 3 = 0
3	3 - 3 = 0
3	3 - 3 = 0
4	4 - 3 = 1
4	4 - 3 = 1
6	6 - 3 = 3

Measuring Variability: The Standard Deviation

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How to calculate standard deviation, s_x :

- 1) Find the mean of the distribution.
- 2) Calculate the *deviation* of each value from the mean.
- 3) Square each of the deviations.

$$\bar{x} = \frac{1 + 2 + 2 + 2 + 3 + 3 + 3 + 3 + 4 + 4 + 6}{11} = 3$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	$1 - 3 = -2$	$(-2)^2 = 4$
2	$2 - 3 = -1$	$(-1)^2 = 1$
2	$2 - 3 = -1$	$(-1)^2 = 1$
2	$2 - 3 = -1$	$(-1)^2 = 1$
3	$3 - 3 = 0$	$0^2 = 0$
3	$3 - 3 = 0$	$0^2 = 0$
3	$3 - 3 = 0$	$0^2 = 0$
3	$3 - 3 = 0$	$0^2 = 0$
4	$4 - 3 = 1$	$1^2 = 1$
4	$4 - 3 = 1$	$1^2 = 1$
6	$6 - 3 = 3$	$3^2 = 9$
		Sum = 18

Measuring Variability: The Standard Deviation

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How to calculate standard deviation, s_x :

- 1) Find the mean of the distribution.
- 2) Calculate the *deviation* of each value from the mean.
- 3) Square each of the deviations.
- 4) Add all the squared deviations, divide by $n - 1$.

$$s_x^2 = \frac{18}{11-1} = 1.80$$

This value is known as the **sample variance**. (In this case, the units would be "squared close friends.")

Measuring Variability: The Standard Deviation

Eleven high school students were asked how many "close" friends they have. Here are their responses: 1 2 2 2 3 3 3 3 4 4 6

How to calculate standard deviation, s_x :

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- 3) Square each of the deviations.
- 4) Add all the squared deviations, divide by $n - 1$.
- 5) Take the square root.

$$s_x = \sqrt{\frac{18}{11-1}} = 1.34 \text{ close friends}$$

This value is known as the **sample standard deviation**.

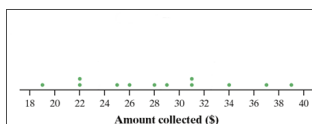
Measuring Variability: The Standard Deviation

Properties of Standard Deviation

- s_x is always greater than or equal to 0.
- Larger values of s_x indicate greater variation.
- s_x is not a resistant measure of variability.
- s_x measures variation about the mean.

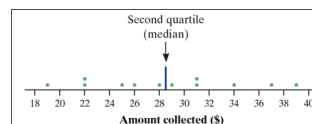
Measuring Variability: The Interquartile Range (IQR)

The **quartiles** of a distribution divide the ordered data set into four groups having roughly the same number of values. To find the quartiles, arrange the data values from smallest to largest and find the median.



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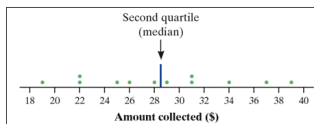


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The **first quartile Q_1** is the median of the data values that are to the left of the median in the ordered list.

The **third quartile Q_3** is the median of the data values that are to the right of the median in the ordered list.

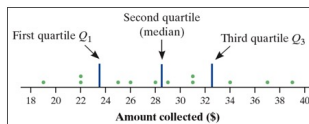


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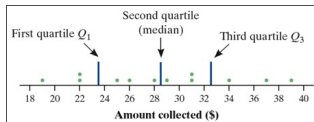


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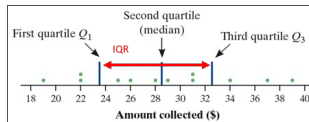
The **interquartile range (IQR)** is the distance between the first and third quartiles of a distribution. In symbols: $IQR = Q_3 - Q_1$

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The **interquartile range (IQR)** is the distance between the first and third quartiles of a distribution. In symbols: $IQR = Q_3 - Q_1$

Measuring Variability: The Interquartile Range (IQR)

Travel times for 20 New Yorkers:

10 | 30 | 5 | 25 | 40 | 20 | 10 | 15 | 30 | 20 | 15 | 20 | 85 | 15 | 65 | 15 | 60 | 60 | 40 | 45

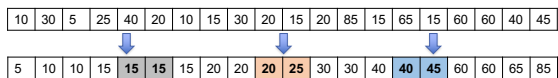
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10 | 30 | 5 | 25 | 40 | 20 | 10 | 15 | 30 | 20 | 15 | 20 | 85 | 15 | 65 | 15 | 60 | 60 | 40 | 45
5 | 10 | 10 | 15 | 15 | 15 | 20 | 20 | 20 | 25 | 30 | 30 | 40 | 40 | 45 | 60 | 60 | 65 | 85

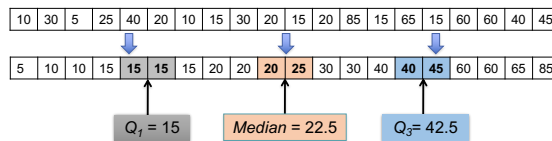
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Travel times for 20 New Yorkers:



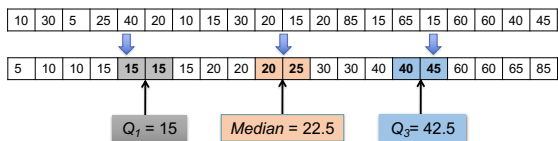
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Travel times for 20 New Yorkers:



Measuring Variability: The Interquartile Range (IQR)

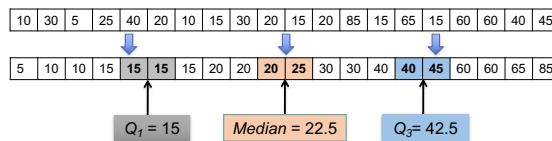
Travel times for 20 New Yorkers:



$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 42.5 - 15 \\ &= 27.5 \text{ minutes} \end{aligned}$$

Measuring Variability: The Interquartile Range (IQR)

Travel times for 20 New Yorkers:



$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 42.5 - 15 \\ &= 27.5 \text{ minutes} \end{aligned}$$

Interpretation: The range of the middle half of travel times for the New Yorkers in the sample is 27.5 minutes.

Identifying Outliers

Although there are several rules for outliers, one of the most common rules is the $1.5 \times IQR$ rule.

HOW TO IDENTIFY OUTLIERS: THE $1.5 \times IQR$ RULE

Call an observation an outlier if it falls more than $1.5 \times IQR$ above the third quartile or below the first quartile. That is,

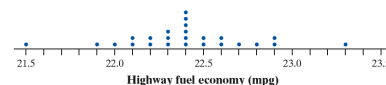
Low outliers $< Q_1 - 1.5 \times IQR$ High outliers $> Q_3 + 1.5 \times IQR$

Identifying Outliers

Highway fuel economy ratings for twenty-five 2018 Toyota 4Runners tested by the EPA:

21.5 21.9 22.0 22.1 22.1 22.2 22.2 22.3 22.3 22.3 22.4 22.4 22.4 22.4 22.4 22.5 22.5 22.6 22.6 22.7 22.8 22.9 22.9 23.3

$Q_1 = 22.2$ mpg
 $Q_3 = 22.6$ mpg
 $IQR = 0.4$ mpg

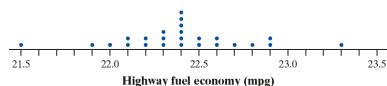


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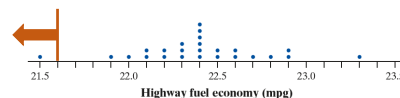
Low outliers $< Q_1 - 1.5 \times IQR = 22.2 - 1.5 \times 0.4 = 21.6$

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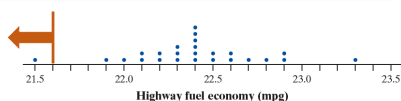
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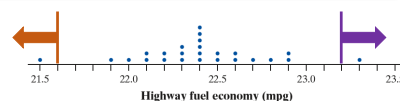
High outliers $> Q_3 + 1.5 \times IQR = 22.6 + 1.5 \times 0.4 = 23.2$

Identifying Outliers

Highway fuel economy ratings for twenty-five 2018 Toyota 4Runners tested by the EPA:

21.5 21.9 22.0 22.1 22.2 22.2 22.3 22.3 22.3 22.4 22.4 22.4
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$Q_1 = 22.2$ mpg
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Low outliers $< Q_1 - 1.5 \times IQR = 22.2 - 1.5 \times 0.4 = 21.6$

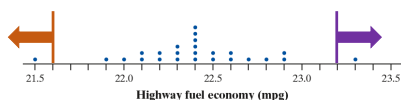
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Identifying Outliers

Highway fuel economy ratings for twenty-five 2018 Toyota 4Runners tested by the EPA:

21.5 21.9 22.0 22.1 22.2 22.2 22.3 22.3 22.3 22.4 22.4 22.4
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$Q_1 = 22.2$ mpg
 $Q_3 = 22.6$ mpg
 $IQR = 0.4$ mpg



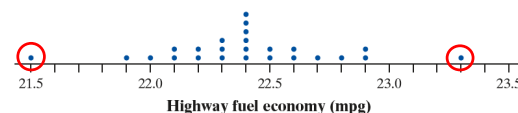
Low outliers $< Q_1 - 1.5 \times IQR = 22.2 - 1.5 \times 0.4 = 21.6$

High outliers $> Q_3 + 1.5 \times IQR = 22.6 + 1.5 \times 0.4 = 23.2$

The cars with fuel economy ratings of 21.5 mpg and 23.3 mpg would be considered outliers by the $1.5 \times IQR$ rule.

Identifying Outliers

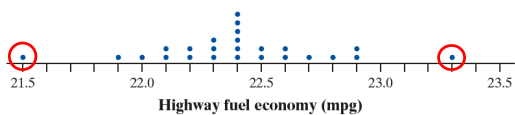
Why look for outliers?



Identifying Outliers

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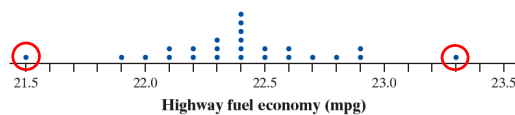
1. They might be inaccurate data values.



Identifying Outliers

Why look for outliers?

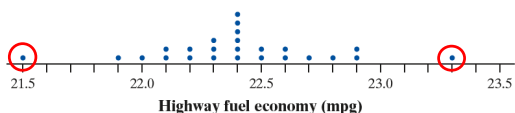
1. They might be inaccurate data values.
2. They can indicate a remarkable occurrence.



Identifying Outliers

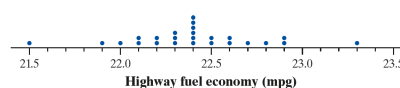
Why look for outliers?

1. They might be inaccurate data values.
2. They can indicate a remarkable occurrence.
3. They can heavily influence the values of some summary statistics, like the mean, range, and standard deviation.



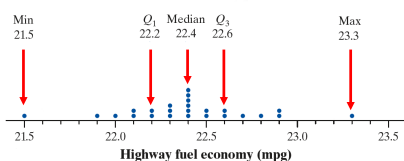
Making and Interpreting Boxplots

The **five-number summary** of a distribution of quantitative data consists of the minimum, the first quartile Q_1 , the median, the third quartile Q_3 , and the maximum.



Making and Interpreting Boxplots

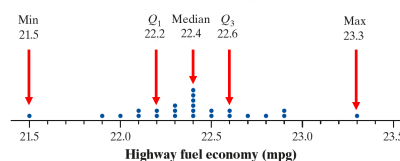
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Making and Interpreting Boxplots

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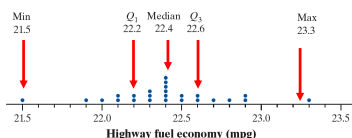
A **boxplot** is a visual representation of the five-number summary.



Making and Interpreting Boxplots

How to Make a Boxplot

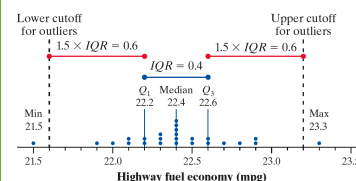
- Find the five-number summary.
- Identify outliers using the $1.5 \times \text{IQR}$ rule.
- Draw and label the horizontal axis.
- Scale the axis.
- Draw a box.
- Mark the median.
- Draw whiskers.



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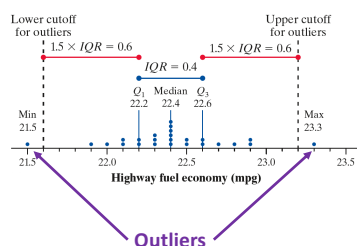
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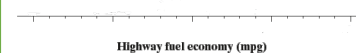
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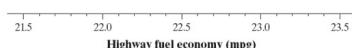
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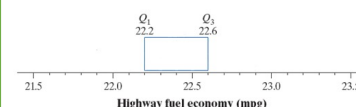
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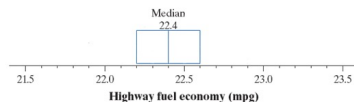
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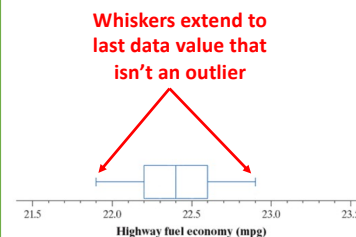
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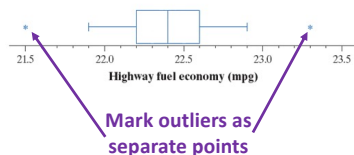
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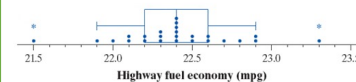
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CAUTION:

- Boxplots do not display each individual value in a distribution.
- Boxplots don't show gaps, clusters, or peaks.



Section Summary

LEARNING TARGETS

After this section, you should be able to:

- ✓ CALCULATE measures of center (mean, median) for a distribution of quantitative data.
- ✓ CALCULATE and INTERPRET measures of variability (range, standard deviation, IQR) for a distribution of quantitative data.
- ✓ EXPLAIN how outliers and skewness affect measures of center and variability.
- ✓ IDENTIFY outliers using the $1.5 \times \text{IQR}$ rule.
- ✓ MAKE and INTERPRET boxplots of quantitative data.
- ✓ Use boxplots and numerical summaries to COMPARE distributions of quantitative data.